



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC — GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CS G525
Course Title : Advanced Computer Networks
Instructor-in-charge : YASH SINHA (yash.sinha[AT]pilani.bits-pilani.ac.in, 6120-J.)
Team of Instructors : TBA

1. Course description: This is a graduate level course on computer networking and assumes a student has a basic understanding of computer network concepts. This course is a topic-based course which primarily covers topics from Internet Architecture, Internet Congestion Control, Software Defined Networking, Delay Tolerant Networks, Wireless Networking, Quality of Service & Traffic Engineering, Network Performance & Management, Overlay Networks and Network Applications. Several research papers/articles/case studies will be discussed on different aspects of computer networking. This course will introduce students to the basic design principles on which today's Internet is based upon along with the current and emerging research topics in the computer networking area. In addition, this course will cover some recent proposals to improve network performance, functionality and scalability. This course will provide plenty of opportunity for learning by doing through projects, assignments and laboratory exercises.
2. Scope and Objective:
 - a. To understand the state of the art in network protocols, network architectures, and networked systems.
 - b. To develop a strong understanding of the core concepts of computer networks.
 - c. To get practice of reading the research papers and critically understanding the research of others.
 - d. To engage ourselves in networking research.
3. Course learning outcomes:
 - CLO1.** Analyze the design principles, architectures, protocols, and performance trade-offs of modern computer networks.
 - CLO2.** Critically evaluate networking research by identifying key contributions, assumptions, limitations, and alternative design choices.
 - CLO3.** Design, implement, and experimentally evaluate solutions to networking problems using appropriate tools and methodologies.

4. Textbooks and references:

There is no prescribed textbook for this course. The course contents will be covered through several research papers. Students can consider following textbooks to refresh their basics on networking:

TB1. Kurose James F and Keith W. Ross: Computer Networking: A Top-Down Approach Featuring the Internet

TB2. L. Peterson and B. Davie, Computer Networks: A Systems Approach

Note: Several technical papers/articles from the literature will be assigned throughout the semester. These will be made available to the students through the course page.

5. Modules

Module	Topics	Learning Outcomes
M1	Internet Architecture & Principles: Overview of network building blocks, Design Philosophy of DARPA Internet, End-to-End Design Principle, layers and protocols, Internet design: Challenges and Solutions	✓ To understand the Internet Design Philosophy and its limitations in today's context. ✓ To learn the pitfalls in existing design and solutions.
M2	Software Defined Networking (SDNs): SDN Architecture and Components, SDN Controller, Network Programmability, Network Function Virtualization (NFV), SDN Frameworks, Use cases for traffic monitoring & management, bandwidth scheduling and monitoring.	✓ To learn about centralized control vs. distributed control of the network. ✓ To understand about network functions, virtualization, and architectural principles of SDNs ✓ To understand technical challenges and potential issues arising in SDN deployment.
M3	Congestion Control & Traffic Management: Congestion control principles, TCP congestion control, Modern Congestion Control Approaches: QUIC, BBR, Multipath and QoS Routing, Traffic Management Algorithms: FIFO, FQ, RED, Codel, etc.	✓ To understand the network traffic control and management mechanisms used for the Internet. ✓ To evaluate traffic management solutions and their applicability in different contexts. ✓ To analyze active queue management algorithms.
M4	Wireless & Mobile Networks: Wireless Networks fundamentals, Wi-Fi Protocol, Mobile IP and Micro Mobility Protocols, TCP protocol Performance Issues in Wireless Networks, Modified TCP for Wireless Networks.	✓ To understand wireless networks and node mobility issues in IP networks and their solutions. ✓ To understand TCP performance challenges in wireless networks. ✓ To learn about solutions for improving TCP performance in wireless networks.
M5	Overlay Networks & Data Center Networks: Overlay Network Design Principles, Motivation, Applications & Protocols: Distributed searching systems based on DHTs, Data Center Networks Architectures, TCP-like Protocol for Data Center Networks	✓ To learn about overlay networks. ✓ Applications-based overlays like content searching, content distribution, filesharing, routing, content caching, etc. ✓ To analyze different architectures for designing Data Center networks ✓ To compare Data Center TCP and TCP protocols.

6. Research Paper / Reading List

M1: Internet Architectures & Principles

1. The Design Philosophy of The DARPA Internet Protocols [Clark 1988]
2. End-to-End Argument in System Design [Saltzer 1984]
3. Tussle in Cyberspace: Defining Tomorrow's Internet [Clark 2005]
4. A Survey of the Research on Future Internet Architectures [Jianli Pan 2011]
5. NDN Project Technical Report [Zhang 2010]

M2: Software Defined Networking

6. Software Defined Networking: The New Norm of Networks, White Paper [ONF 2012]
7. OpenFlow: Enabling Innovation in Campus Networks [N McKeown 2008]
8. Ten Things to Look for in an SDN Controller
9. FlowSense: Monitoring Network Utilization with Zero Measurement Cost [Curtis Yu 2013]
10. P4: Programming Protocol-Independent Packet Processors [Bosshart 2014] Visit: <https://p4.org/>

M3: Congestion Control & Traffic Management

11. Congestion Avoidance and Control [Jacobson 1988]
12. Analysis of Increase and Decrease Algorithms for Congestion Avoidance in Computer Networks [Raj Jain 1989]
13. The QUIC Transport Protocol: Design and Internet-Scale Deployment [Adam 2017]
14. BBR: Congestion-Based Congestion Control [Cardwell 2016]
15. How Hard Can It Be? Designing and Implementing a Deployable Multipath TCP [Raiciu 2012]
16. Random Early Detection Gateways for Congestion Avoidance [Sally Floyd 1993]
17. The BLUE Active Queue Management Algorithms [Feng 2002]
18. Controlling Queuing Delay [Van Jacobson 2012]

M4: Wireless & Mobile Networking

19. A Survey of Wi-Fi 6: Technologies, Advances, and Challenges [Erfan 2022]
20. A Comparison of Mechanisms for Improving TCP Performance over Wireless Links [Balakrishnan 1996]
21. Mobility in IP Networks: From Link Layer to Application Layer Protocols and Architectures [Johnson 2010]

M5: Overlay Networks & Data Center Network

22. Chord: A Scalable Peer-to-peer Lookup Service for Internet Applications [Stoica 2001]
23. A Scalable, Commodity Data Center Network Architecture [Al-Fares 2008]
24. Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network [Arjun Singh 2015]
25. Data Center TCP (DCTCP) [Mohammad Alizadeh 2010]

7. Course Format & Weekly Workflow

The course will follow a research-oriented and discussion-driven format. Each week will combine instructor-led discussion of networking concepts and research papers with critical paper discussions/debates. Students will work in permanent groups, finalized during the first week, for both the Research Paper Debate and the Research Project.

A typical week will follow the structure below:

- Monday & Wednesday: Instructor-led discussion of concepts, research papers, systems, and case studies.
- Friday: Research Paper Debate (Red Team vs. Blue Team), as scheduled.
- Research Paper Debate: The Blue Team will present and defend the assigned work, while the Red Team will critically examine its assumptions, design choices, limitations, and potential research gaps. The instructor will conclude the discussion by synthesizing the key technical takeaways.
- Research Project: Projects will run throughout the semester in parallel with the course. Groups will progressively move through problem formulation, literature review, implementation, experimentation, analysis, and demonstration.
- Hands-on Support: Ungraded hands-on exercises/tutorials using tools such as Mininet, OpenFlow, and POX/Ryu will be provided where appropriate to support project implementation and experimental evaluation.

Instructor-led Class |
 Research Paper Debate |
 Research Project |
 Holiday |
 Exam

Week	Monday	Wednesday	Friday	Remarks
W1 Aug 3–7	 Aug 3 • M1: Course Introduction + <i>The Design Philosophy of The DARPA Internet Protocols</i> [Clark 1988]	 Aug 5 • M1: <i>End-to-End Argument in System Design</i> [Saltzer 1984]	 Aug 7 • M1: End-to-End Argument continued + discussion	 Permanent Debate + Project groups finalized
W2 Aug 10–14	 Aug 10 • M1: <i>Tussle in Cyberspace: Defining Tomorrow's Internet</i> [Clark 2005]	 Aug 12 • M1: <i>A Survey of the Research on Future Internet Architectures</i> [Jianli Pan 2011]	 Aug 14 • Debate 1: Tussle + Future Internet Architectures	 Aug 15: Independence Day (H)
W3 Aug 17–21	 Aug 17 • M1: <i>NDN Project Technical Report</i> [Zhang 2010]	 Aug 19 • M2: <i>Software Defined Networking: The New Norm of Networks</i> [ONF 2012]	 Aug 21 • Debate 2: NDN + SDN	—
W4 Aug 24–28	 Aug 24 • M2: <i>OpenFlow: Enabling Innovation in Campus Networks</i> [N Mckeown 2008]	 Aug 26: Milad-un-Nabi (H) Extra Class: • M2: OpenFlow continued + Mininet	 Aug 28: Raksha Bandhan (H) Extra Class • Debate 3: OpenFlow	 Project Proposal – 5% due Aug 28 Extra classes for holidays
W5 Aug 31–Sep 4	 Aug 31 • M2: <i>Ten Things to Look for in an SDN Controller</i>	 Sep 2 • M2: <i>FlowSense</i> [Curtis Yu 2013]	 Sep 4: Janmashtami (H) • Extra Class • Debate 4: SDN Controllers FlowSense	Extra classes for holidays
W6 Sep 7–11	 Sep 7	 Sep 9 • M2: P4 continued	 Sep 11 • Debate 5: P4	—

	• M2: P4: Programming Protocol-Independent Packet Processors [Bosshart 2014]			
W7 Sep 14–18	 Sep 14: Ganesh Chaturthi (H) • Extra Class • M3: Congestion Avoidance and Control [Jacobson 1988]	 Sep 16 • M3: Analysis of Increase and Decrease Algorithms [Raj Jain 1989]	 Sep 18 • Debate 6: Jacobson + Jain	Extra classes for holidays
W8 Sep 21–25	 Sep 21 • M3: The QUIC Transport Protocol [Adam 2017]	 Sep 23 • M3: BBR: Congestion-Based Congestion Control [Cardwell 2016]	 Sep 25 • Debate 7: QUIC + BBR	—
W9 Sep 28 – Oct 2	 Sep 28 • M3: How Hard Can It Be? Designing and Implementing a Deployable Multipath TCP [Raiciu 2012]	 Sep 30: Progress Review I – 5%	 Oct 2: Gandhi Jayanti (H)	 Meetings for Progress Review I
Oct 5–10	 Mid-Semester Examinations	 Classwork Suspended	 Oct 9: Mid-Semester Examination – 20%	—
W10 Oct 12–16	 Oct 12 • M3: Random Early Detection Gateways for Congestion Avoidance [Sally Floyd 1993]	 Oct 14 • M3: BLUE [Feng 2002] + <i>Controlling Queuing Delay</i> [Van Jacobson 2012]	 Oct 16 • Debate 8: RED vs BLUE / CoDel	—
W11 Oct 19–23	 Oct 19 • M4: A Survey of Wi-Fi 6 [Erfan 2022]	 Oct 21 • M4: A Comparison of Mechanisms for Improving TCP Performance over Wireless Links [Balakrishanan 1996]	 Oct 23 • Debate 9: Wi-Fi 6 + TCP over Wireless	 Oct 20: Dussehra (H)
W12 Oct 26–30	 Oct 26 • M4: Mobility in IP Networks [Johnson 2010]	 Oct 28 • M5: Overlay Networks	 Oct 30: Progress Review II – 5%	—
W13 Nov 2–6	 Nov 2 • M5: Chord: A Scalable Peer-to-peer Lookup Service for Internet Applications [Stoica 2001]	 Nov 4 • M5: A Scalable, Commodity Data Center Network Architecture [Al-Fares 2008]	 Nov 6 • Debate 10: Chord + Data Center Network Architecture	—
W14 Nov 9–13	 Research Project Consultation	 Research Project Consultation	 Research Project Consultation	 Diwali • Nov 8: Deepavali/Govardhan Puja (H)
W15 Nov 16–20	 Nov 16 • M5: Jupiter Rising [Arjun Singh 2015]	 Nov 18 • M5: Data Center TCP (DCTCP) [Mohammad Alizadeh 2010]	 Nov 20: Project Presentation with Demo/Viva – I	 Final Work & Report – 15% due Nov 20, 12 noon.
W16 Nov 23–27	 Nov 23 • M5: DCTCP / Data Center Networks continued	 Nov 25: Course Synthesis + Compre Discussion	 Nov 27: Project Presentation with Demo/Viva – II	 Nov 24: Guru Nanak Jayanti (H) •

8. Evaluation Scheme:

Component	Weightage	Duration	Date and Time	Comments
Research Paper Debate	10%	In class	Throughout semester	• Blue Team: Present and defend the paper • Red Team: Critique assumptions, limitations and identify research gaps • Rest of class: Participate in discussion and vote on teams' performance
Research Project	35%	—	Throughout semester	Details given below
Mid-Semester Examination	20%	90 min	09/10 FN2	Individual • Concepts, analysis and problem solving • Closed Book
Comprehensive Examination	35%	180 min	13/12 AN	Individual • Concepts, design, problem solving and critical analysis • Closed Book
Total	100%			

Research Project Proposal	Weightage	Duration	Date and Time	Comments
	5%	—	28 Aug 2026 (Fri)	Problem formulation, related work, proposed methodology and evaluation plan
Progress Review I	5%	—	30 Sep 2026 (Wed)	Problem refinement • Setup • Baseline implementation • Initial progress
Progress Review II	5%	—	30 Oct 2026 (Fri)	Working implementation • Experiments • Preliminary results • Feedback incorporation
Final Work & Report	15%	—	20 Nov 2026 (Fri)	Implementation, experiments, analysis and final technical report
Presentation with Demo/Viva	5%	15 min	20 & 27 Nov 2026 (Fri)	Project demonstration and viva; individual understanding assessed
Total	35%			

All students are required to install the following tools Tools/Software in their laptops to perform the lab exercises, assignment/project work.

- Protocol performance analysis and Software Defined Network simulation using Mininet - Openflow – POX/Ryu Controller

Criterion for NC: [30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.]

9. Attendance Policy: **Each student is expected to attend all classes and participate in all evaluation components.**

10. Chamber consultation hours: Wed 3:00 PM – 4:00 PM

11. Makeup Policy: Makeup will be given only for genuine cases (accompanied by proof) for Mid-semester or comprehensive exam only in genuine cases as per the CSIS department makeup policy guidelines. In all cases prior permission from the Instructor-In-Charge is must.

12. Notices: Notices regarding the course will be uploaded ONLY on the LMS NALANDA

Instructor-in-charge
CS G525